Johnson Center Donor Advised Fund Report Research

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Introduction

Donor Advised Funds (DAFs) are charitable checking accounts. Donors can choose to contribute assets to a DAF held by a charitable organization in exchange for some tax benefits, and they can advise when and where those funds are distributed. DAFs have risen in prominence over recent years, and charities are interested in knowing how much is held in these accounts so they can make a prediction for how much they're going to receive from these funds.

While the information about every organization's DAFs is publicly available, there's a long delay, typically around eighteen months, between when the IRS receives the information about what's held in these accounts and when they actually release that information. Some organizations release that information on their own, or are willing to disclose it to the Johnson Center, but around one quarter of organizations' financial information is not released in time for the annual DAF fund report. Thus, the Johnson Center is tasked with estimating that missing data.

Our group worked on evaluating and improving the methods used to estimate unreleased DAF data. We used the following variables from data publicly available from the Johnson Center:

- **Accounts**: the total quantity of DAFs held by an organization at the end of the each tax year
- **Contributions**: the total monetary contributions to all DAF accounts held by an organization at the end of the each tax year
- **Grants**: the total monetary amount distributed from DAF accounts held by an organization to charitable recipients at the end of the each tax year

- **Assets**: the total value of funds held in DAF accounts held by an organization at the end of the each tax year
- **Type/Subtype**: the category of charitable organization (e.g., national charities, single-issue charities, universities, hospitals, and other nonprofits)

The following questions guided our research: First, how accurate have previous predictions made by the Johnson Center been? Second, how do these predictions hold up across different charity types since 2008? Third, is there a better way to make predictions for the Donor Advised Fund report?

With clear goals and raw data that needed to be cleaned prior to our research and analysis, we began our work.

Data Cleaning and Combining

We received two different spreadsheets from the Johnson Center, one with historical data that they inherited, and one with more recent data curated by the Johnson Center.

The historical spreadsheet was, frankly, a mess. It had a horizontal format where each charity was its own row, and each column represented each of the above variables as well as the year that it represents. For example, there would be a column for accounts in 2007, a column for accounts in 2008, and so on for every year since 2007. One of the other problems was that this data had been collected by multiple different organizations and was therefore labeled differently every few years.

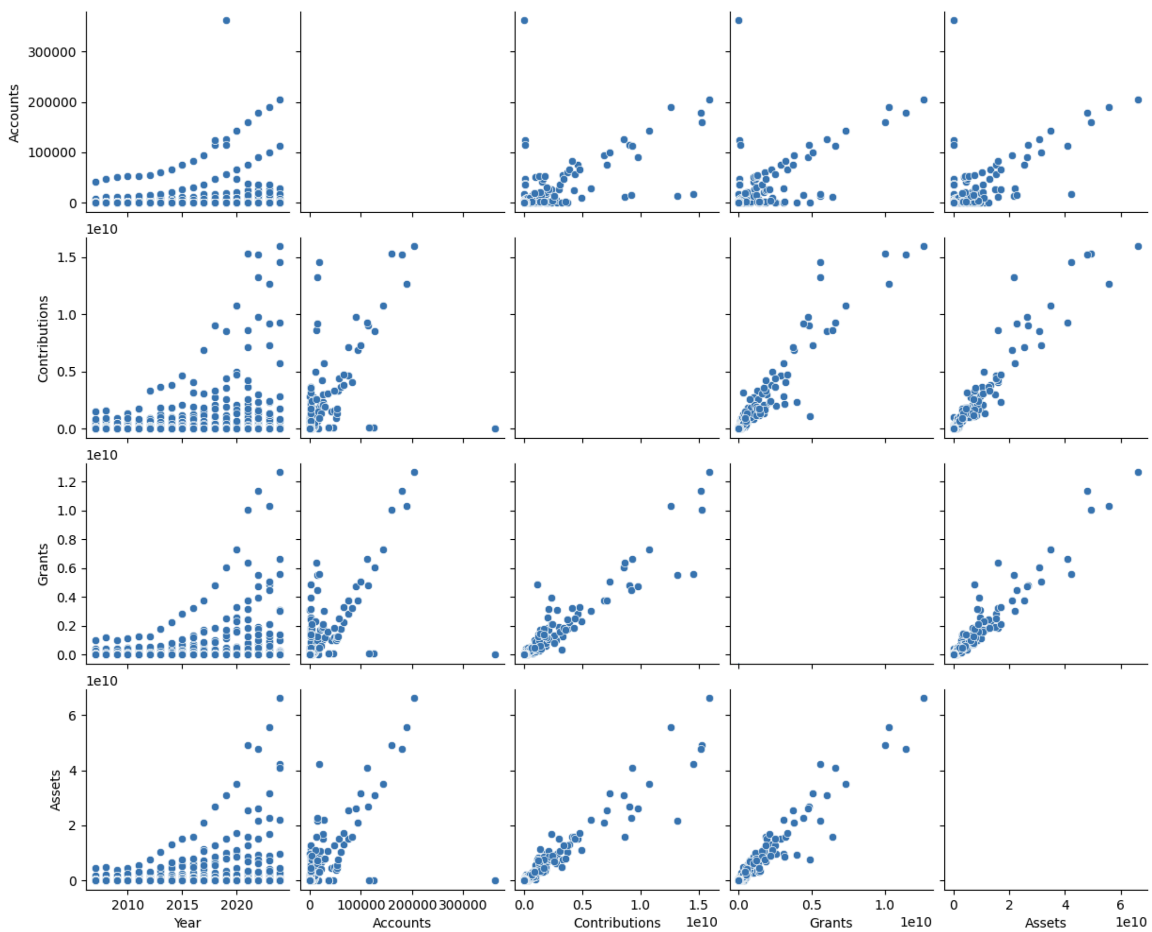
The second spreadsheet, with the more recent data, had a beautiful vertical format and data divided up by charity and year. Additionally, this file contained a “subtype” column that was absent in the other file. Note that this document only contained data from 2019 to 2024. One of our original goals was to combine these mismatched files to create a vertically formatted dataset containing both the historical data and more recent data.

Of the many columns of our datasets, we only wanted to keep eight. Those columns were "ein" (employer identification number), "year", "accounts", "contributions", "grants", "assets", "type", and "subtype". After that, we created a new dataset, which we will henceforth refer to as the vertical historic dataset. It was formed by relabeling and reassigning columns from the historical data to fit the format of the vertical file, and dropping the columns that we wanted to get rid of. We then concatenated files from the vertical recent data and the vertical historic data to form the combined dataset we were looking for. Pending some final data cleaning, our data was formatted for analysis with a total of 9 columns and 23329 rows. However, the data itself still had some issues.

Not all of the charities considered in our data have been around since 2007. Because of this, and for many other reasons, there were many nan values in our datasets. We removed all of these charity/year combinations with nan values from our data. We also encountered negative values in our monetary columns and removed those as well. No charity should realistically have negative monetary values in their accounts. Following the data cleaning, we had 8066 rows of data, representing about 35% of the data.

Initial Data Analysis

We investigated the correlation between various variables to decide what relationships may be worth investigating when finding better imputation methods and to locate outliers. For example, the graph below features a scatter plot with every combination of each numeric variable, and year, on both the x and y axes. In the cases when it plots a variable against itself, it leaves those sections blank. It’s also worth noting that the scale for contributions, grants, and assets is 10^{10} because we’re working with big numbers. All of these graphs have a generally positive correlation, with larger funds generally staying larger throughout the years. We also observed an interesting phenomenon where some funds will have a smaller number of accounts but a larger amount of contributions, grants, and assets in those accounts.



The Johnson Weighted Average Method

The Johnson Center imputes missing data for its annual report using a weighted average method based on group-level growth rates. When information is missing and cannot be obtained before the report is released, values are predicted using the prior year's data. The data is first grouped by type and subtype, where each type–subtype combination defines a group. Within each group, the available data is grouped by year to compute totals for accounts, contributions, assets, and grants. These group-level totals are then ordered chronologically, and growth rates are calculated separately for each metric using consecutive years only. Growth rates are calculated separately for each group and each metric using the formula:

$$Growth\ Rate\ =\ \frac{Current\ Year\ Total - Prior\ Year\ Total}{Prior\ Year\ Total}$$

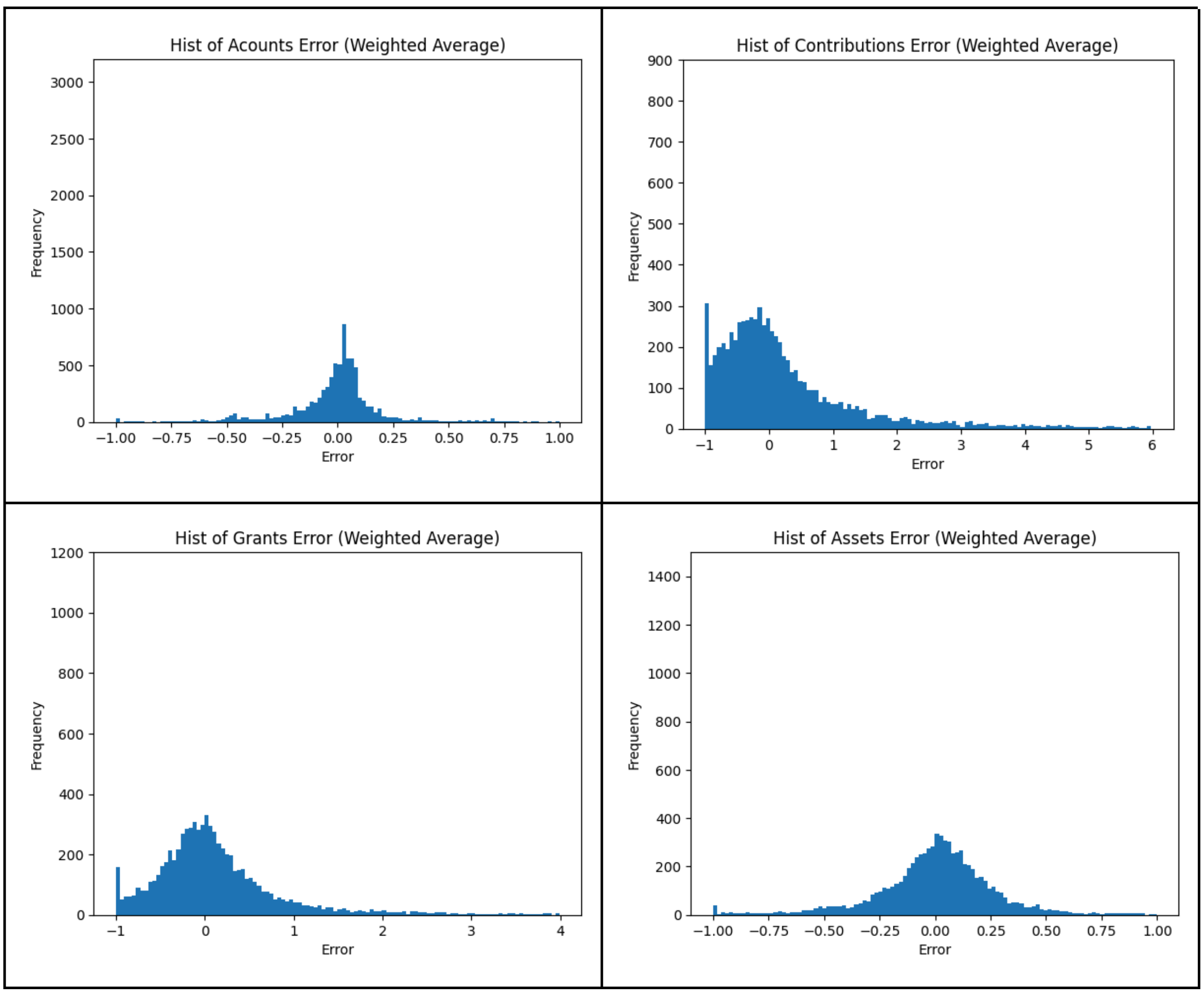
Growth rates are only retained when the group has observations in consecutive years, and non-consecutive changes are excluded to avoid distortion. These group-level growth rates are then merged back into the original dataset. Finally, for each organization, predicted values are generated by applying the corresponding group growth rate to the prior year's actual value, following the formula: predicted value = prior year value × (1 + growth rate). Predictions are only produced when the organization itself has consecutive-year data, ensuring consistency and reliability in the imputation process.

To calculate error, we found the relative error of the predictions made using the Johnson Center's imputation method. We compared the actual known data to the predicted data to see how close the predictions were. The average relative error and the median relative error are reported below:

Category	**Average Relative Error**	**Median Relative Error**
Accounts	21.70%	7.26%
Contributions	357.41%	54.53%
Grants	231.34%	34.16%
Assets	64.42%	13.90%

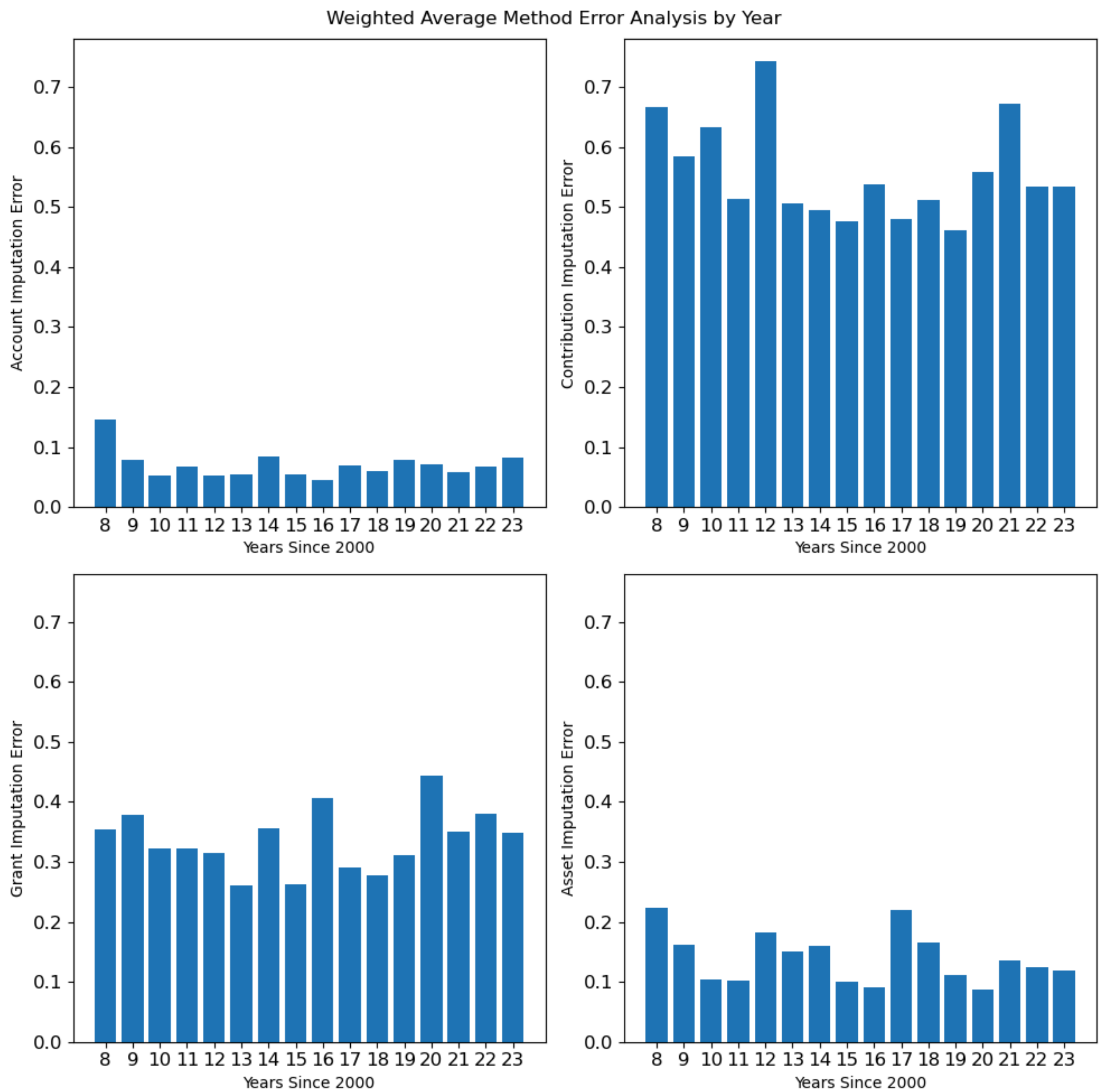
It's unsurprising that the average relative error is high, since it's easily influenced by outliers.

We plotted the actual error in histograms for each category to visualize the spread of the prediction errors.



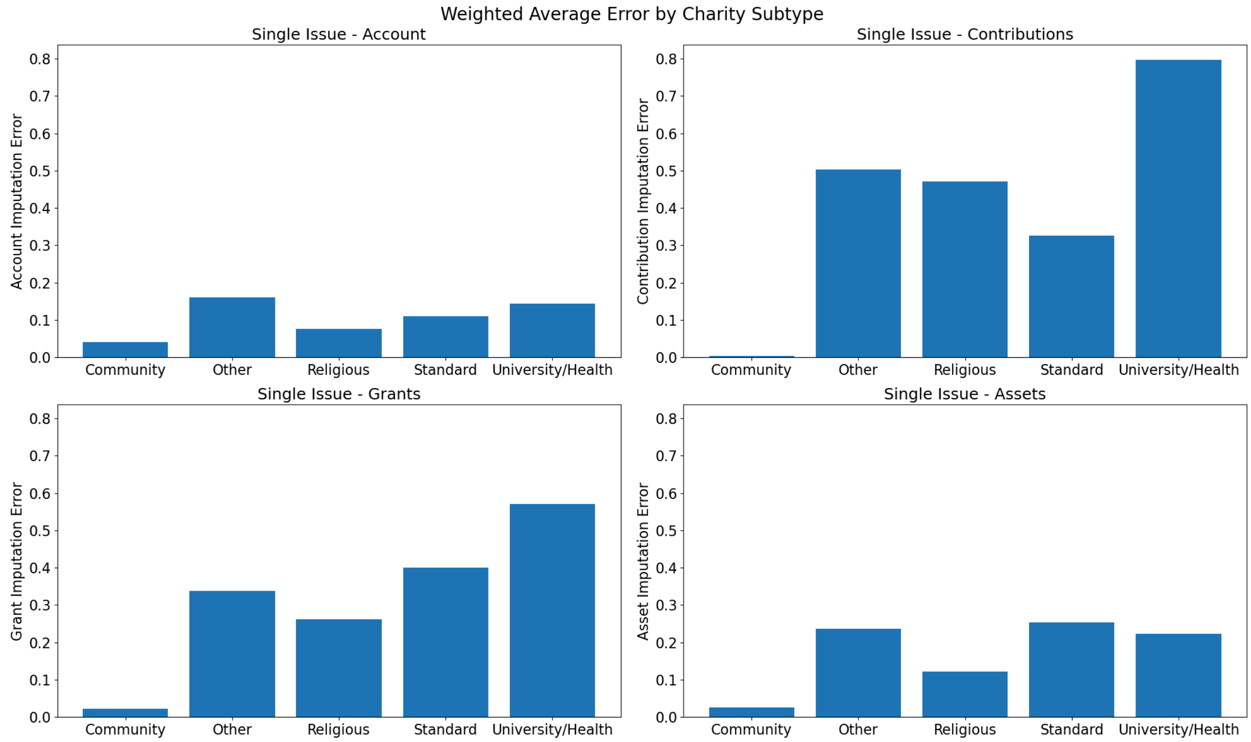
The weighted average method shows notably different error distributions across the four metrics. Accounts error is tightly concentrated near zero, suggesting reasonably accurate predictions for that metric. However, contributions and grants errors are broadly spread and right-skewed, with long tails extending to +6 and +4 respectively, indicating a systematic tendency to underpredict large values. Assets error is also fairly tight but shows a slight left skew, meaning the model occasionally overpredicts asset values significantly.

But how do these errors change over time? From looking at the weighted average imputation error in our predictions of the four monetary variables over time, there doesn't seem to be any particular pattern for when the error is spiking. In fact, when looking at the graphs below, each variable error seems to spike in different years. So, despite the presence of spikes, this method seems relatively time independent, indicating a resistance to time sensitive factors.



We also aimed to analyze the error according to each type and subtype of charity. As we can see from the plots below, charities of subtype "Other" or "Universities and Healthcare" have higher error than the other subtypes of charities. Note that those plots are only representative of the "Single Issue" charity type, which is the most prominent charity type. However, there are significantly less of these charities than of the other subtypes. Thus, it makes sense that we would see higher error in these categories because there was less data to make predictions off of.

Likewise, there are many charities with the subtype "Community," so it makes sense that the error is so low.



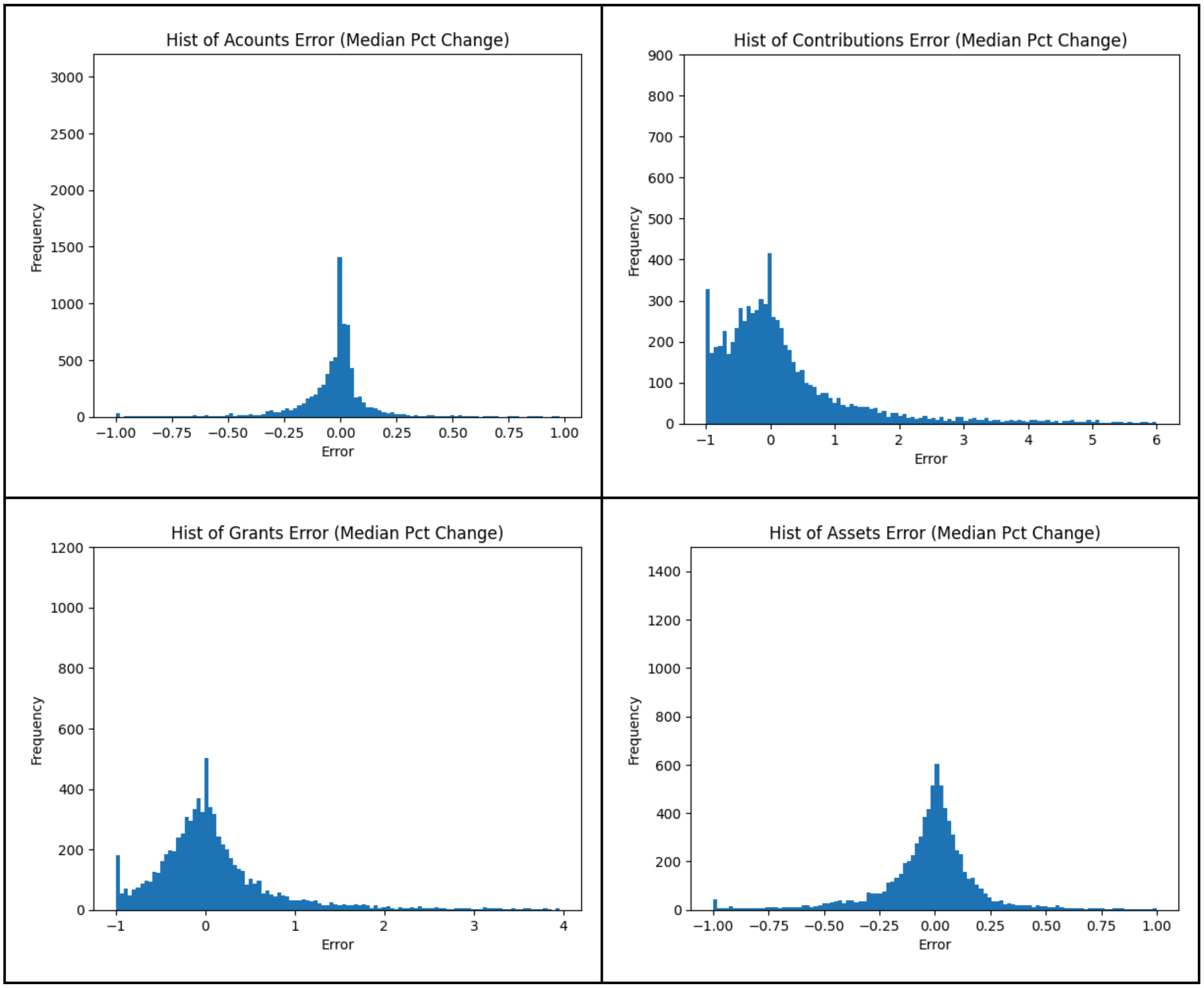
Median method

As an initial attempt to improve upon the Johnson Center's imputation approach, we developed a method based on median growth rates derived from organization-level trends within each type–subtype group. The dataset is first sorted by organization (EIN) and year to ensure proper chronological order, and the difference in years is calculated to identify consecutive observations. For each metric—accounts, contributions, grants, and assets—the year-over-year percent change is computed at the organization level; however, only changes from consecutive years are retained, while non-consecutive changes are excluded to avoid distortion. These individual growth rates are then grouped by year, type, and subtype, and the median growth rate is calculated for each metric within each group. The median is used instead of the mean to reduce the influence of outliers. These group-level median growth rates are then merged back into the original dataset. Finally, predicted values are generated by multiplying the prior year's actual value by one plus the corresponding median growth rate, following the formula: predicted value = prior year value × (1 + median growth rate). This produces imputed values for accounts, contributions, grants, and assets when current-year data is missing.

To calculate error, we found the relative error of the predictions made using the median percent change imputation method. We compared the actual known data to the predicted data to see how close the predictions were. The average relative error and the median relative error are reported below:

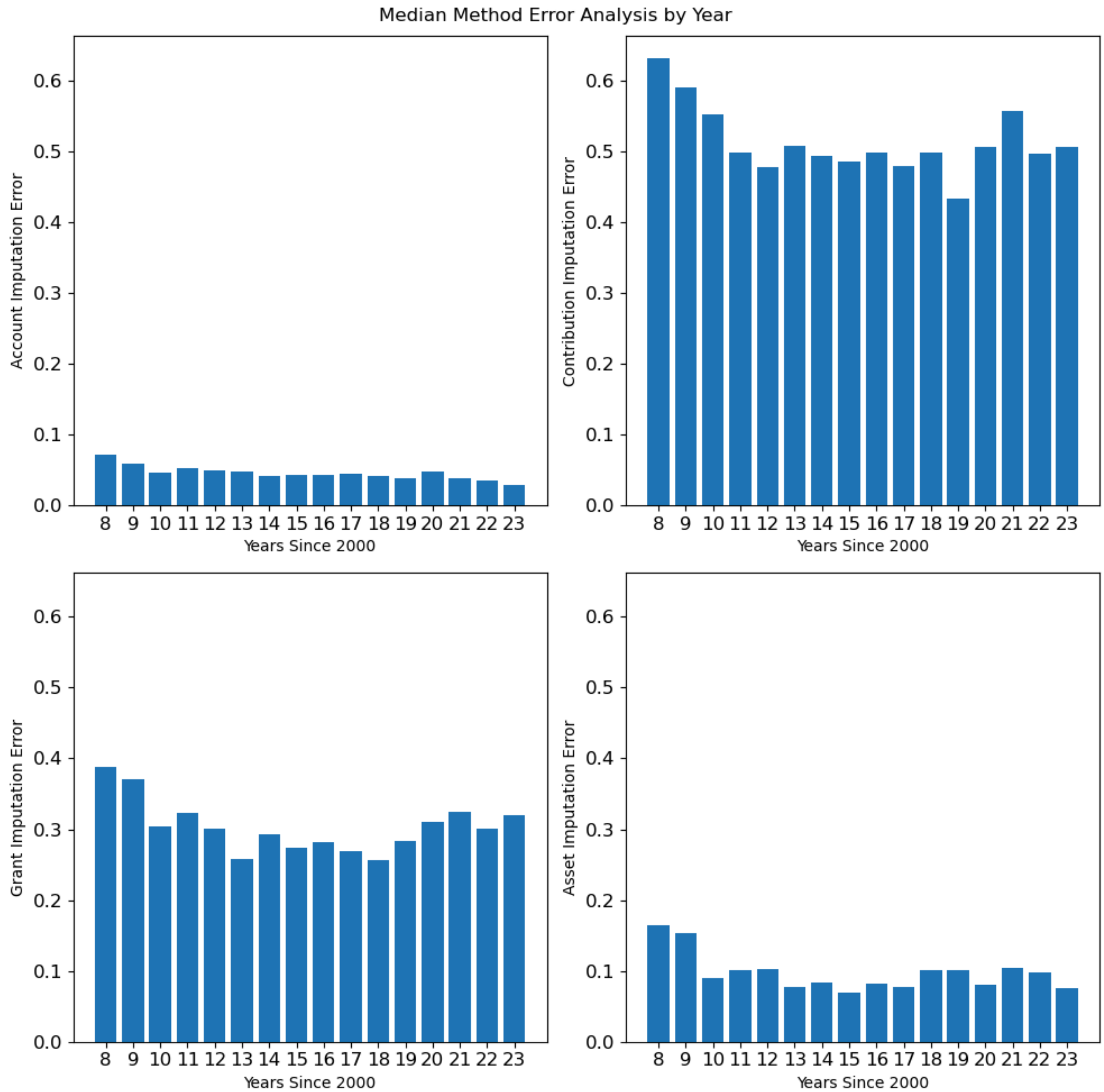
Category	**Average Relative Error**	**Median Relative Error**
Accounts	13.08%	4.38%
Contributions	339.10%	49.51%
Grants	262.74%	29.46%
Assets	60.25%	8.83%

We plotted the actual error in histograms for each category to visualize the spread of the prediction errors.

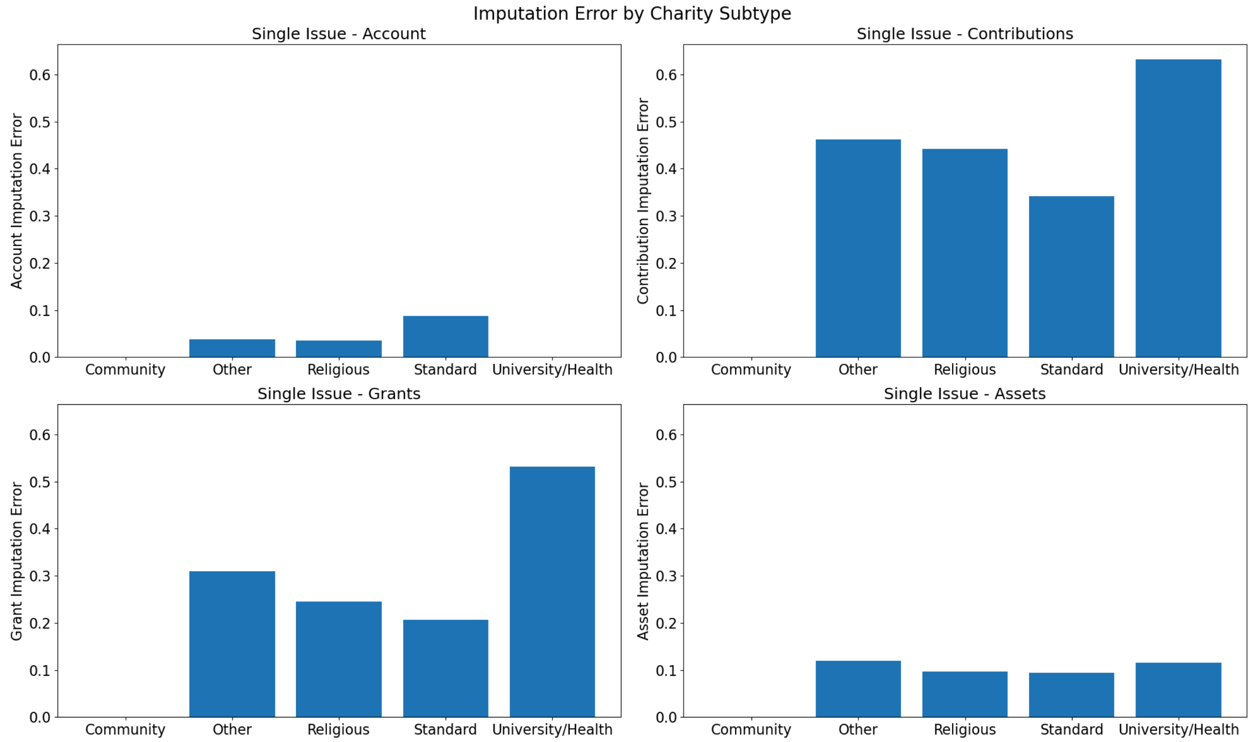


The median percent change method produces error distributions that look qualitatively similar to the weighted average, but with some differences in concentration. Accounts error remains tightly clustered near zero, though with a slightly sharper peak than the weighted average version. Contributions and grants again display wide, right-skewed distributions with long positive tails, reflecting the same underprediction tendency for larger values. Assets error is more symmetric and tighter than in the weighted average method, suggesting this approach performs somewhat better for that particular metric.

When considering median method imputation error over time, we again observed that the spikes in the plots are not aligned by year. However, these graphs seem to suggest a general gradual improvement to imputation methods since 2008. There are certainly many spikes of error in between 2008 and 2023 but by 2023, there is a noticeable decrease in error.



When investigating the change in error based on type, we find similar results to the weighted average error method explored previously. The one difference in these plots is that the "Universities and Healthcare" subtype has zero or near zero error in accounts.



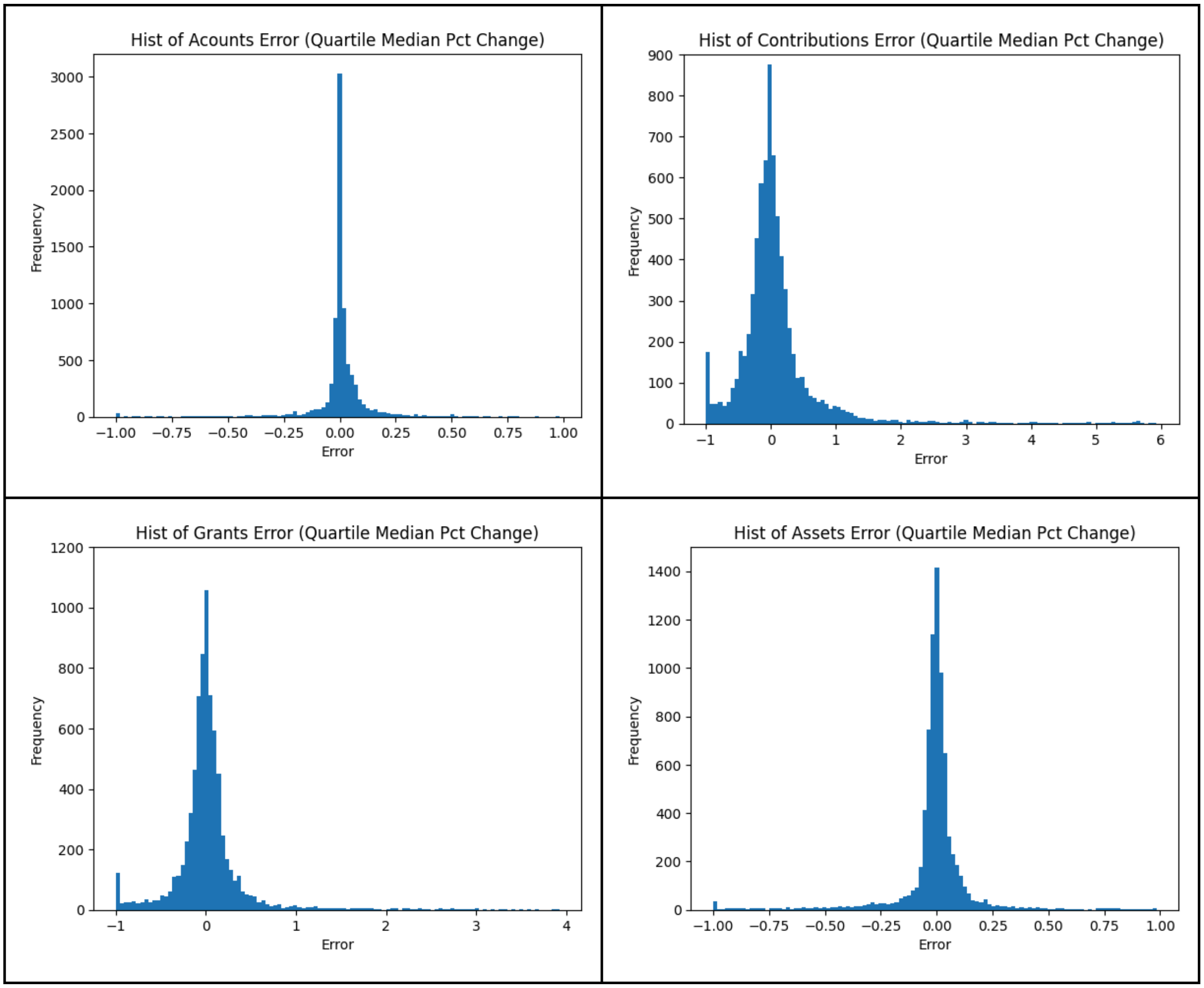
Split Quartile Median Method

As a further refinement of our imputation approach, we developed the split quartile median method, which builds on the median growth framework by segmenting organizations into growth-based brackets within each type–subtype group. The dataset is first sorted by organization (EIN) and year, and year-over-year percent changes are calculated for each metric—accounts, contributions, grants, and assets—using only consecutive years, with non-consecutive changes excluded to avoid distortion. Within each year, type, and subtype group, organizations are then divided into four quartile-based brackets based on their percent change for each metric, representing relative growth performance (bottom 25% through top 25%). If there are too few observations to form quartiles, a default middle bracket is assigned to maintain stability. After assigning these brackets, the data is grouped again by year, type, subtype, and growth bracket, and the median growth rate is computed separately for each metric within each bracket. These bracket-specific median growth rates are then merged back into the dataset. Finally, predicted values are generated by applying the corresponding bracket-level median growth rate to each organization's prior year value, following the formula: predicted value = prior year value × (1 + median growth rate). By incorporating growth-based segmentation, this method aims to produce more tailored and accurate imputations by accounting for differences in growth behavior across organizations within the same group

To calculate error, we found the relative error of the predictions made using the split quartile median imputation method. We compared the actual known data to the predicted data to see how close the predictions were. The average relative error and the median relative error are reported below:

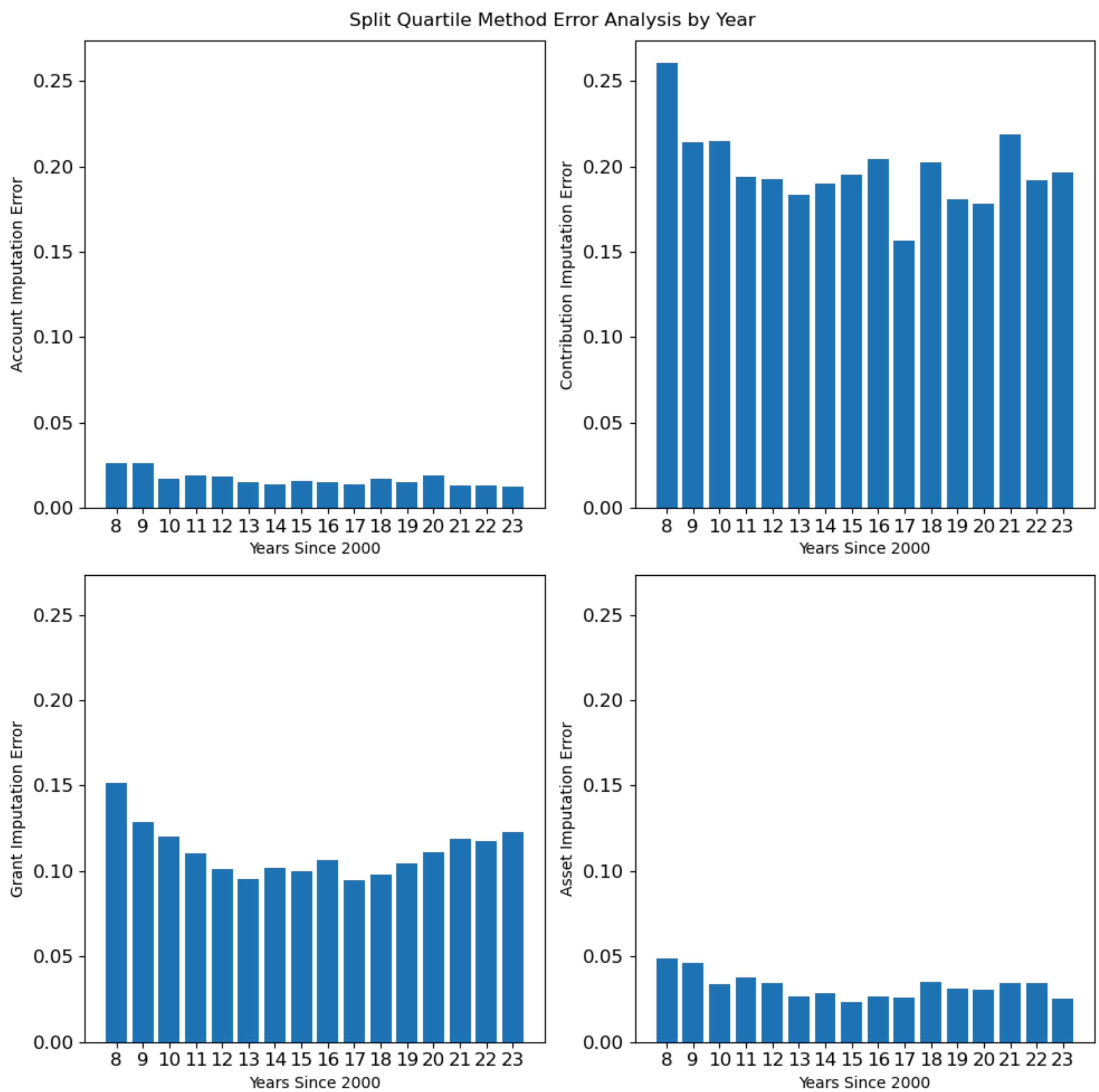
Category	**Average Relative Error**	**Median Relative Error**
Accounts	9.15%	1.60%
Contributions	92.10%	19.59%
Grants	100.90%	10.90%
Assets	44.80%	3.17%

We plotted the actual error in histograms for each category to visualize the spread of the prediction errors.

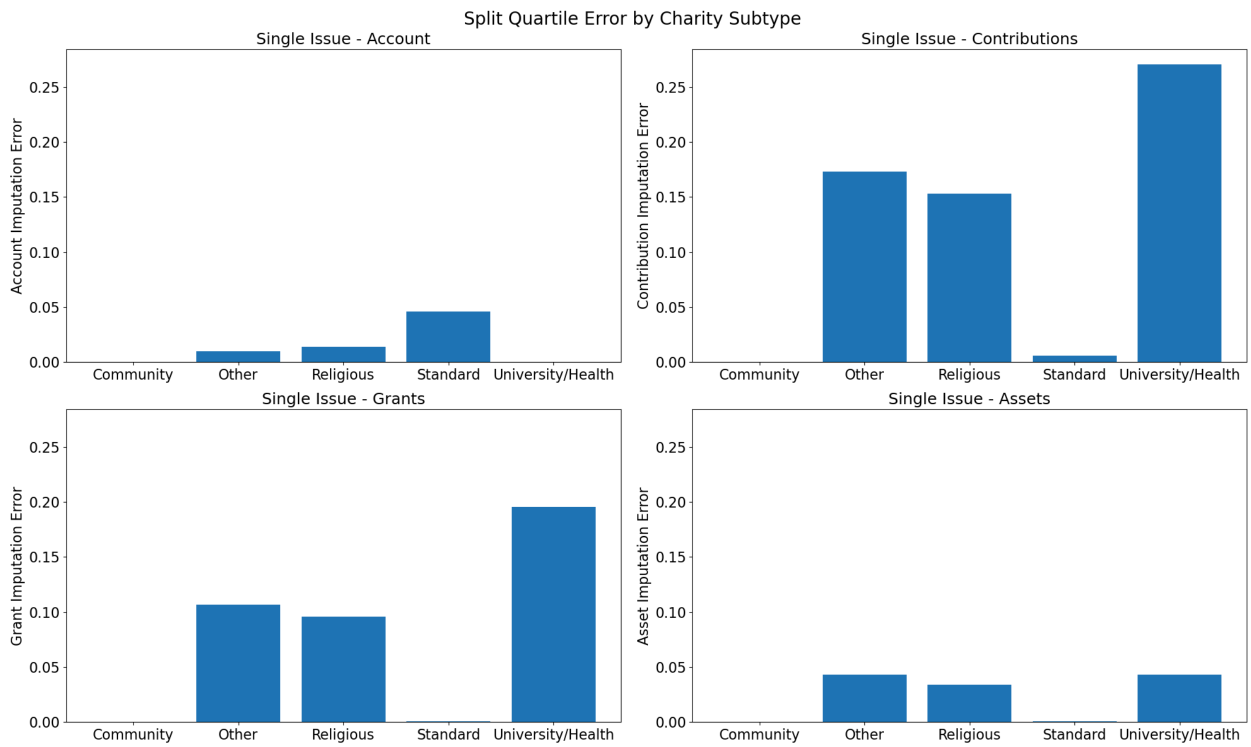


The split quartile median percent change method produces the most well-behaved error distributions of the three approaches across all four metrics. All four histograms show dramatically tighter, more symmetric peaks centered near zero, with most predictions falling close to the actual values. Contributions and grants still retain some right skew and occasional large positive errors, but the improvement in concentration is substantial compared to the other two methods. Overall, this method appears to generate the most consistent and least biased predictions, making it the strongest performer of the three.

Compared to the median method, the split quartile median method imputation error does not appear to have any different trends. Again, we find misaligned spikes that suggest a lack of heavy temporal impact on our errors. However, we do note a general decrease in error since 2008, as we noted in the median method imputation error graphs.



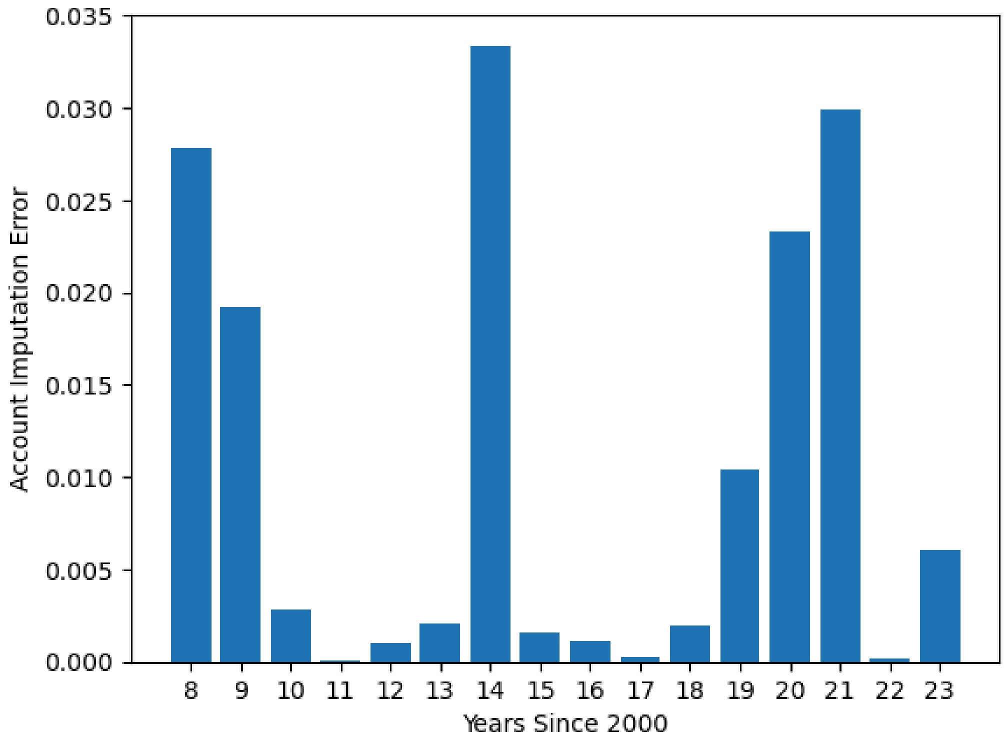
Yet again, we notice a general trend of higher error amongst “Other” and “Universities and Healthcare” charity subtypes. The “Standard” charity subtype has reacted differently to the quartering of the data than the other subtypes. Strangely, it reports the highest account error of all of the subtypes, but near zero errors for all other categories. We are unsure why this is the case, but it is an outlier amongst the other conclusions we have made.



Aggregate Error

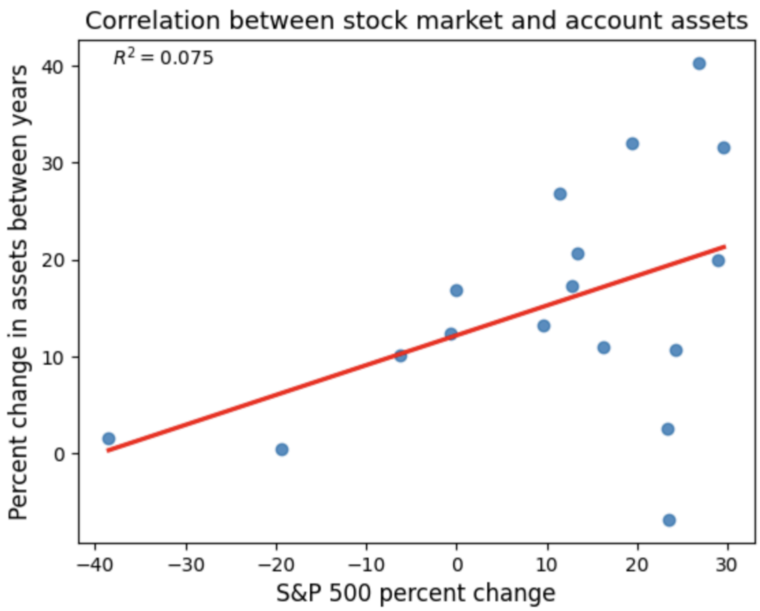
We originally found the total number of accounts in all charities for a given year before computing the error. This placed all of the individual accounts into one group, where we then found the error for the entire group. While it gave us very small error values, we realized that it's most important to consider the error at the individual charity level, so we switched our error method. However, before we decided to make this change, we had an interesting finding. When computing the account imputation error for every year that we had data for (2008-2023), we found that there were large spikes related to economic recessions and recoveries, such as the 2008 economic recession, the 2014 economic recovery, and economic uncertainty and

COVID-19 in 2021. The state of the economy is something that is difficult to account for in a mathematical model, but it may be a way to create an improved imputation method in the future.



Suboptimal Linear Regression Imputation Method

First, we wanted to investigate the relationship between the stock market and the percent change in the assets of accounts for each year using linear regression. Since the organizations holding these assets are likely to have a good portion of the money invested in the stock market and thus, the S&P 500 in an attempt to grow their assets, we thought that it was an avenue worth pursuing.



There's a positive correlation between S&P500 growth and the growth in the total amount of money in the accounts over the years, which meant that we wanted to look into using it as a potential weight when making our linear regressions.

There was a slight positive correlation between S&P 500 growth and the yearly change in the assets held, but the R^2 value was very low. However, we still thought that it was an interesting avenue to explore, so we tried using it as a weight for linear regressions. Sadly but unsurprisingly, once we actually started trying to create linear regressions, we found that our error was much higher than our current method, and that using the percent change of the stock market actually made the error worse. Still optimistic, albeit less so than when we first started creating our regression, we decided to try one more approach.

We winsorized the data, meaning that instead of directly removing outliers, we changed the values of the top and bottom n% of data to the bottom or top value within their respective bin. To give an example, in a 90% Winsorization (5% on each tail), all values below the 5th percentile are set to the 5th percentile, and all values above the 95th percentile are set to the 95th percentile. We also tried throwing out the top and bottom n% of the data altogether. While both of these methods did marginally improve our linear regressions, the errors were still much higher than the ones from our split quartile median method, so we didn't pursue this further.

Conclusion

Over the course of this project, we investigated each of the research questions given to us by the Johnson Center. Let's revisit those questions.

1. How accurate have previous predictions of unreleased DAF data been compared to reported actual values?

The Johnson center's original method had a median relative error of 7.26% for accounts, 54.53% for contributions, 34.16% for grants, and 13.90% for assets. Contributions and grants are known to be variable, so seeing high errors was not unexpected, especially given the large outliers in our data. When investigating the individual errors of the charities, we observed a wide distribution of errors on our histogram, suggesting that this method has high errors for many charities.

2. How does prediction accuracy vary across different types of charitable organizations over time since 2008?

We observed that the charity types with the least data were the ones with the most relative error. The subtypes of "other" and "healthcare" for example, had very infrequent data, and we observe higher errors across each quantitative variable for those subtypes. This is good because the focus of the model should be the bigger charity types and subtypes.

We also observed that prediction accuracy varied when economic stress hit. When looking at aggregate error, the errors spike in the years 2008, 2014, 2021. These were the years of economic recession, economic growth, and COVID uncertainty. As for non-aggregate error, we only saw a gradual decrease in error from 2008 to 2023.

3. Can methods be found and employed that improve the accuracy of these predictions beyond the methods currently used?

We found our split quartile median method to have much smaller average and median relative error across all four quantitative variables than the Johnson Center method. We suspect there are ways to further improve this calculation that could be discovered upon further study, such as looking into different or potentially more bins to split up the data. However, we currently recommend using the split quartile median method when predicting DAF financial data for future years.